Skeleta in Complex Projective Line

Yusuf Berk AKÇAY

9 October 2026

Quantum-mechanical motivation

The sphere $S^2 \cong \mathbb{CP}^1$ is the classical phase space of a spin with fixed magnitude: a point specifies its direction.

In holomorphic geometric quantization, at level k,

$$\mathcal{H}_k = H^0(\mathbb{CP}^1, \mathcal{O}(k)) \cong \{a_0 + a_1 z + \cdots + a_k z^k\}, \qquad \dim \mathcal{H}_k = k + 1.$$

Thus a polynomial represents a quantum state. Nonzero scalar multiples represent the same physical pure state.

This is the spin-$j = k/2$ representation. Degrees $1, 2, 3$ correspond to spin $\frac{1}{2}, 1, \frac{3}{2}$.

We use the standard holomorphic quantization without the half-form correction.

Skeleta and the semiclassical viewpoint

Semiclassical analysis asks how quantum states acquire a geometric description as $k \to \infty$, with effective $\hbar \sim k^{-1}$.

For a section s, its squared norm $F = \|s\|^2$ gives a landscape on the classical sphere. Its gradient flow selects the skeleton Γ_s.

A concrete example: the coherent state $p(z) = z^k$ concentrates near the pole ∞ as k grows. Its skeleton is precisely that point.

Motivating question: what geometry does Γ_s reveal about the state s?

A skeleton is associated to a quantum state. Identifying a general skeleton with a semiclassical state requires further assumptions.

Complex Manifolds

Let M be a topological space which is Hausdorff and second countable. We say M is a complex manifold of dimension n provided that there exists an open cover $\{U_\alpha\}_{\alpha \in I}$ of M, where each U_α is called a coordinate chart, and a family of homeomorphisms $\phi_\alpha : U_\alpha \to \phi_\alpha(U_\alpha) \subseteq \mathbb{C}^n$ such that for all $\alpha \neq \beta$ in I, the transition maps

$$\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \to \phi_\beta(U_\alpha \cap U_\beta)$$

are holomorphic functions between open subsets of $\mathbb{C}^n$.

Main Example:

$$\mathbb{CP}^1 = \{\ell \leq \mathbb{C}^2 \,|\, \ell \text{ is a one dimensional } \mathbb{C}\text{-subspace}\} = \{[X : Y] \,|\, (X, Y) \in \mathbb{C}^2 - \{0\}\}/\sim$$

where $\sim$ is defined as $[X : Y] \sim [\lambda X : \lambda Y]$ for $0 \neq \lambda \in \mathbb{C}$. One can endow $\mathbb{CP}^1$ with the quotient topology and get a Hausdorff, second countable topological space.

Coordinate Charts on $\mathbb{CP}^1$

To turn $\mathbb{CP}^1$ into a complex manifold of dimension one, define two coordinate charts on $\mathbb{CP}^1$ as

$$\mathbb{C}_z = \{[z:1] \,|\, z \in \mathbb{C}\} \simeq \mathbb{C} \qquad \mathbb{C}_w = \{[1:w] \,|\, w \in \mathbb{C}\} \simeq \mathbb{C}$$

Clearly, these two charts cover $\mathbb{CP}^1$, and in their intersection $\mathbb{C}_z \cap \mathbb{C}_w = \mathbb{C}_z - \{[0:1]\} = \mathbb{C}_w - \{[1:0]\}$ the transition map is defined by the relation

$$[z:1] \sim [1:w] \iff z = \frac{1}{w}$$

As it is common in projective geometry, by an abuse of notation we will denote the point $w = 0$ as $z = \infty$ and vice versa. Then, we have $\mathbb{CP}^1 \simeq S^2$ the Riemann sphere, where one should think of $\mathbb{C}_z$ as the 'northern hemisphere' and the $\mathbb{C}_w$ as the 'southern hemisphere'.

The Riemannian and Symplectic Manifold Structure of $\mathbb{CP}^1$

As we discussed, the complex projective line $\mathbb{CP}^1$ is diffeomorphic to the usual unit sphere S^2, in particular it admits a metric coming from the restriction of the Euclidean metric of the $\mathbb{R}^3$ to the submanifold $S^2 \subseteq \mathbb{R}^3$. This metric can be expressed as

$$ds^2 = \frac{4(dx^2 + dy^2)}{(1 + x^2 + y^2)^2} = \frac{4|dz|^2}{(1 + |z|^2)^2}$$

where in the first expression we use the coordinates $(x, y) = (\Re z, \Im z)$ and in the second expression we used the coordinates $(z, \bar{z}) = (x + iy, x - iy)$. From these expressions, we can obtain the area form on $\mathbb{CP}^1$ as

$$\omega = \frac{4\, dx \wedge dy}{(1 + x^2 + y^2)^2} = \frac{2i\, dz \wedge d\bar{z}}{(1 + |z|^2)^2}$$

This 2-form ω has the following properties:

- $d\omega = 0$
- ω provides a nondegenerate bilinear form on each tangent space $T_p\mathbb{CP}^1$ where $p \in \mathbb{CP}^1$.

Such a 2-form on a manifold M is called a symplectic form, and the pair (M, ω) is called a symplectic manifold. Hence, when endowed with the area form ω, the complex projective line $\mathbb{CP}^1$ becomes a symplectic manifold.

Holomorphic Line Bundles

A holomorphic line bundle on a complex manifold M, called the base space, is a complex manifold L, called the total space, together with a surjective holomorphic projection map $\pi : L \to M$ such that

- Each fiber $\pi^{-1}(p) =: L_p$, $p \in M$, has a one dimensional complex vector structure, *i.e.* linearly isomorphic to $\mathbb{C}$.
- There exists an open cover $\{U_\alpha\}_{\alpha \in I}$ of M such that one has the holomorphic diffeomorphism

$$\psi_\alpha : \pi^{-1}(U_\alpha) \simeq U_\alpha \times \mathbb{C}$$

  such that restricting this diffeomorphism to a fiber $\pi^{-1}(p)$ gives the point $p \in U_\alpha$ in the first slot and a linear isomorphism $\pi^{-1}(p) \simeq \mathbb{C}$ in the second slot.
- On the intersections $U_\alpha \cap U_\beta$, we have two diffeomorphisms

$$\psi_\alpha : \pi^{-1}(U_\alpha \cap U_\beta) \simeq (U_\alpha \cap U_\beta) \times \mathbb{C} \qquad \psi_\beta : \pi^{-1}(U_\alpha \cap U_\beta) \simeq (U_\alpha \cap U_\beta) \times \mathbb{C}$$

  Now, the maps $\psi_\alpha \circ \psi_\beta^{-1} : (U_\alpha \cap U_\beta) \times \mathbb{C} \to (U_\alpha \cap U_\beta) \times \mathbb{C}$ has the form $(x, v) \mapsto (x, g_{\alpha\beta}(x)v)$ where the maps $g_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(\mathbb{C}) \simeq \mathbb{C}^\times$ are called the transition functions. These transition functions satisfy the cocycle property: On triple intersections $U_\alpha \cap U_\beta \cap U_\gamma$ we have $g_{\beta\gamma} g_{\alpha\beta} = g_{\alpha\gamma}$.

Holomorphic Line Bundles over $\mathbb{CP}^1$

There is a complete classification of line bundles over $\mathbb{CP}^1$. To state this classification, first define the tautological line bundle τ as

$$\tau = \{(\ell, v) \in \mathbb{CP}^1 \times \mathbb{C}^2 \mid v \in \ell\}$$

with the projection $\pi : \tau \to \mathbb{CP}^1$ defined as $\pi(\ell, v) = \ell$. Now, for the coordinate charts $\mathbb{C}_z$ and $\mathbb{C}_w$ of $\mathbb{CP}^1$, we have the trivialization maps

$$\psi_z([z:1], (v_1, v_2)) = ([z:1], v_2) \qquad \psi_w([1:w], (v_1, v_2)) = ([1:w], v_1)$$

On the intersection $\mathbb{C}_z \cap \mathbb{C}_w$, we have

$$\psi_w \circ \psi_z^{-1}([z:1], v) = \psi_w([z:1], (zv, v)) = ([1:w], zv)$$

Hence, the transition function is given as $g_{wz}(z) = z = 1/w$. Now, we will define a family of line bundles $\mathcal{O}(k)$, $k \in \mathbb{Z}$. First, define $\mathcal{O}(0) = \mathbb{CP}^1 \times \mathbb{C}$ the trivial line bundle, then define $\mathcal{O}(-1) = \tau$ the tautological line bundle, its dual $\mathcal{O}(1) = \tau^\vee$ and

$$\mathcal{O}(k) = \begin{cases} \mathcal{O}(1)^{\otimes k} & k > 0 \\ \mathcal{O}(-1)^{\otimes |k|} & k < 0 \end{cases}$$

Sections of $\mathcal{O}(k)$

Let us give the definition of a section of a line bundle. For a line bundle $\pi : L \to M$, a section is a holomorphic map $s : M \to L$ such that for all $p \in M$, $s(p) \in L$ lies in the fiber of p. The $\mathbb{C}-$space of sections of a line bundle L is denoted as $\Gamma(L)$. Now, let s be a section of $\mathcal{O}(k)$ for $k > 0$. Locally, we can express s in the coordinate charts $\mathbb{C}_z$ and $\mathbb{C}_w$ as

$$\psi_z(s|_{\mathbb{C}_z}) = \left(z, \sum_{n=0}^{\infty} a_n z^n\right) \qquad \psi_w(s|_{\mathbb{C}_w}) = \left(w, \sum_{n=0}^{\infty} b_n w^n\right)$$

The transition relation of $\mathcal{O}(k)$ implies

$$\sum_{n=0}^{\infty} b_n w^n = z^{-k} \sum_{n=0}^{\infty} a_n z^n$$

and equating the coefficients shows that $a_n = b_n = 0$ for $n > k$. Hence, the sections of $\mathcal{O}(k)$ are given by the polynomials of degree at most k in the chart $\mathbb{C}_z$:

$$\Gamma(\mathcal{O}(k)) = \{a_0 + a_1 z + \cdots + a_k z^k \mid a_i \in \mathbb{C}\}$$

Hermitian Structures on Line Bundles

Let $\pi : L \to M$ be a holomorphic line bundle, then a Hermitian structure on L is a smoothly varying family of inner products

$$\langle -, - \rangle_p : L_p \times L_p \to \mathbb{C}$$

such that for all $v, w \in L_p$ and $\alpha \in \mathbb{C}$ we have

- $\langle v, \alpha w \rangle_p = \alpha \langle v, w \rangle_p$
- $\langle v, w \rangle_p = \overline{\langle w, v \rangle_p}$
- $\langle v, v \rangle_p \in \mathbb{R}^{>0}$ for $v \neq 0$

The inner product gives us a norm $||v||^2 := \langle v, v \rangle_p$.

For the tautological line bundle $\tau = \mathcal{O}(-1)$ of $\mathbb{CP}^1$, we can define a Hermitian structure as

$$\langle v, w \rangle_p = \overline{v}_1 w_1 + \overline{v}_2 w_2$$

Then, we get the usual norm $||v||^2 = |v_1|^2 + |v_2|^2$. Using this norm on τ, we can define a norm on the sections of $\mathcal{O}(k)$, *i.e.* to the polynomials $p(z)$ of degree $\leq k$. This norm is given as

$$||s||^2 = \frac{|p(z)|^2}{(1 + |z|^2)^k}$$

The norm becomes a height function

Write $F = \|s\|^2$. For a degree-k homogeneous section,

$$F([X : Y]) = \frac{|P(X, Y)|^2}{(|X|^2 + |Y|^2)^k}, \qquad F(z) = \frac{|p(z)|^2}{(1 + |z|^2)^k}.$$

Rescaling (X, Y) by λ multiplies numerator and denominator by the same factor $|\lambda|^{2k}$.

So F is a well-defined smooth landscape on the sphere. Its zeros are exactly the roots.

Steepest descent on the sphere

A particle at q moves in the direction in which F decreases fastest:

$$\dot{q}(t) = -\nabla_g F(q(t)).$$

The chain rule gives

$$\frac{d}{dt} F(q(t)) = -\|\nabla_g F(q(t))\|_g^2 \leq 0.$$

Thus a nonstationary trajectory always moves downhill and cannot form a periodic orbit.

A **critical point** satisfies $\nabla_g F = 0$. It is a stationary particle: a minimum, a maximum, a saddle, or a degenerate point.

The skeleton is the set that does not escape

Remove small neighborhoods of the roots by choosing a small regular value $\eta > 0$ and keeping

$$W_\eta = \{F \geq \eta\}.$$

A descending particle can cross the boundary and escape toward a root.

$$\Gamma = \{q \in W_\eta : \text{the forward trajectory of } q \text{ stays in } W_\eta\}.$$

With finitely many critical points: retain the non-root critical points and all trajectories descending to them.

The roots themselves are punctures of the surface, not vertices of this skeleton.

Morse and Morse–Smale

Morse: every critical point is nondegenerate. The Hessian has no zero eigenvalue. Near a critical point, suitable coordinates give

$$\begin{array}{ccc} x^2+y^2 & -x^2-y^2 & x^2-y^2 \\ \text{minimum} & \text{maximum} & \text{saddle} \end{array}$$

up to adding the critical value.

Morse–Smale: the flow also has transverse intersections of stable and unstable manifolds. At an intersection, their tangent spaces together span the tangent space of the surface.

For a Morse gradient flow on a surface, this means there are *no saddle-to-saddle trajectories*.

A function can be Morse even when its gradient flow is not Morse–Smale.

Skeleta in degrees one and two

Degree 1
One simple root



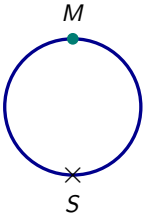
Point

Degree 2
A double root



Point

Degree 2
Two non-antipodal roots



Separatrix loop

Degree 2
Two antipodal roots



Circle of maxima

• Maximum × Saddle Roots are omitted. Pictures are schematic.

The moduli space of cubics

A point of $\mathcal{M}_3$ represents an unordered triple of roots on the sphere, up to a common rotation. Repeated roots are allowed.

Multiplying P by $c \neq 0$ multiplies F by $|c|^2$. It changes the speed of the gradient flow, but not its trajectories.

Generically, three sphere points supply 6 real parameters and rotations remove 3. Thus the moduli space is three-dimensional.

Two different spheres: the root sphere is S^2. The full parameter space is $\mathcal{M}_3 \cong S^3$. A point of the latter describes a whole landscape on the former.

Why only $(M, S) = (1, 2)$ or $(2, 3)$?

For a **Morse cubic with three distinct roots**, let M count maxima and S count saddles. The three roots are the only minima.

The sphere supplies the counting rule:

$$\#\text{minima} - \#\text{saddles} + \#\text{maxima} = \chi(S^2) = 2.$$

Hence $3 - S + M = 2$, or $S = M + 1$.

Being cubic supplies an upper bound: put a maximum at infinity. The remaining critical-point equations reduce generically to a quartic, so there are at most four finite solutions. Thus $M + S \leq 5$. The same bound extends to every Morse cubic.

$$M \geq 1, \qquad 2M + 1 \leq 5 \qquad \Longrightarrow \qquad \boxed{(M, S) = (1, 2) \text{ or } (2, 3)}.$$

Where the special cubics live

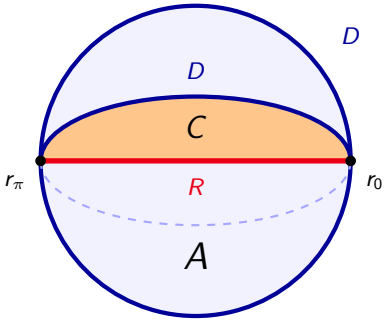
Set	Meaning of one parameter point
A	Morse cubic with one maximum and two saddles
B	Morse cubic with two maxima and three saddles
$C \subset A$	A trajectory connects the two ordinary saddles
D	Distinct roots, but a non-root critical point is degenerate
R	At least two roots coincide

The first two rows count ordinary critical points. A point of C still has a Morse landscape, but its flow is not Morse–Smale.

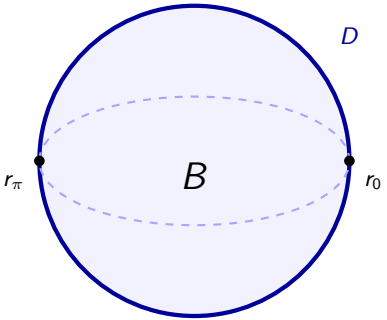
D records degenerate critical points. R records root collisions.

The moduli space: a simplified topological model

H_A: the side containing A



H_B: the side containing B



Glue the entire boundary spheres point by point: $\mathcal{M}_3 \cong S^3$**.**

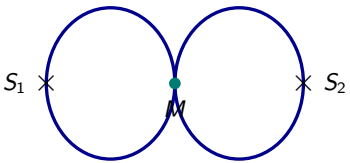
D: the common boundary sphere minus r_0, r_π. R: the collision interval, including its endpoints.

$A = \text{int}(H_A) \setminus R$, $B = \text{int}(H_B)$. $C \subset A$: the interior of the orange disk.

Cubic skeleta: types A, B and C

Type A, off C

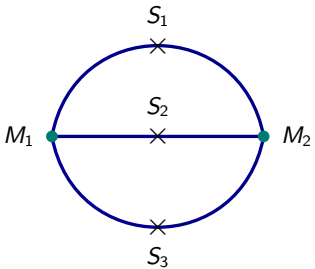
$A \setminus C$



One maximum, two saddles
Morse–Smale

Type B

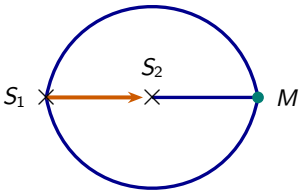
B



Two maxima, three saddles
Morse–Smale

Type C

$C \subset A$



One maximum, two saddles
Morse, not Morse–Smale

• Maximum × Saddle Arrow: descending saddle connection $S_1 \to S_2$